MF-Mamba: Multiscale Convolution and Mamba Fusion Model for Semantic Segmentation of Remote Sensing Imagery

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***Abstract*—Semantic segmentation of remote sensing imagery plays an important role in applications such as environmental monitoring and disaster response. However, challenges such as complex spatial patterns of variable target objects, significant scale variations, and high interclass similarity challenge accurate segmentation. Most existing methods based on convolutional neural networks (CNNs) and Transformers face limitations in modeling multiscale global–local dependencies or often incur high computational costs. Therefore, we propose a multiscale convolution and Mamba fusion model (MF-Mamba) that integrates a CNN encoder with a Mamba-based decoder. The decoder incorporates a global–local state-space (GLSS) module with eight-directional selective scanning mechanisms and multikernel parallel convolutions to capture the rich global–local context. To enhance multiscale feature representation, we developed a channel–spatial attention and dense multiscale feature fusion (CSDF) module, which combines channel–spatial attention and atrous convolutions for multiscale feature fusion. Additionally, a multiscale lateral connection is developed to align encoder features for efficient integration. Experiments on the datasets of ISPRS Vaihingen, ISPRS Potsdam, and the Wuhan Dense Labeling Dataset (WHDLD) demonstrate the superior performance of MF-Mamba compared to existing state-of-the-art methods. It achieves Mean $F1$ scores of 86.71%, 90.70%, and 77.07%. The code is available at https://github.com/Mango-Mars/MF-Mamba**

***Index Terms*—Deep learning, multiscale fusion, remote sensing, semantic segmentation, visual state-space model (SSM).**

I. Introduction

THE rapid advancements in satellite and UAV technologies have made high-resolution remote sensing a vital tool for modern Earth observation, with growing applications in diverse fields such as urban planning, disaster response, and agricultural management [1], [2], [3]. Semantic segmentation plays a key role in these applications by assigning a semantic label to each pixel in an image, thereby accurately interpreting the spatial distribution of land cover types. However, segmentation of fine-scale remote sensing images remains highly challenging due to their inherent characteristics, including complex backgrounds, spatial heterogeneity, and class imbalance [4], [5], [6].

Traditional semantic segmentation approaches typically rely on rule-based classification and conventional machine learning approaches such as random forests [7], conditional random fields [8], or support vector machines [9]. Although these traditional methods have performed well in specific applications, their reliance on manually engineered features and expert-defined rules limits their applicability in complex and heterogeneous remote sensing scenes and in large-area applications in changing environments. Rapid advances in deep learning have increasingly replaced traditional approaches with convolutional neural networks (CNNs), which leverage sliding convolution operations to extract semantic features directly from images. This enables end-to-end prediction while greatly improving segmentation accuracy and efficiency.

As the pioneering CNN architecture, the fully convolutional network (FCN) [10] replaces traditional fully connected layers with convolutional layers, enabling end-to-end pixel-level classification. However, FCN's straightforward upsampling often results in coarse segmentation outputs. To address this, encoder–decoder architectures such as Unet [11] and BiSANet [12] were introduced, where the encoder captures high-level semantic features through progressive downsampling, and the decoder restores spatial resolution via interpolation or transposed convolutions. To further enhance semantic representation and mitigate spatial information loss, various improved architectures have been proposed, incorporating multiscale context and attention mechanisms, including BEDSN [13], Deeplabv3+ [14], and STFF-GA [15]. Although CNNs have become the backbone of many image processing tasks [16], [17], [18], [19], their reliance on small receptive fields due to limited kernel sizes constrains their ability to capture long-range dependencies (LRDs). This limitation is especially problematic in remote sensing, where scenes are often large and complex, containing objects of varying sizes and backgrounds with rich heterogeneity. As a result, CNNs struggle to model the global context effectively, which in turn affects the overall segmentation performance [20].

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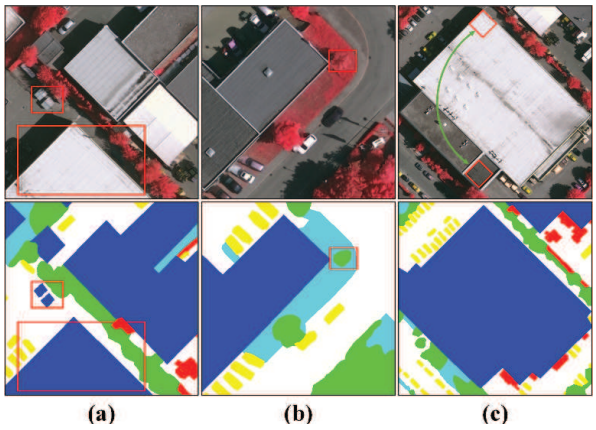


Fig. 1. Examples of typical challenges in remote sensing image semantic segmentation. (a) Scale variation, i.e., small and large structures of the same target object. (b) Spectral similarity across classes, i.e., tree and low vegetation feature similar spectral behavior. (c) High intraclass variability, i.e., significant spectral variations appear in this case within the class "building."

To mitigate the limitations of CNNs, particularly their restricted receptive field, researchers are exploring the application of Transformer architectures to semantic segmentation tasks [21], [22]. Unlike CNNs, the Transformer architecture utilizes a multihead self-attention, which can model long-distance dependencies and capture global context. Hybrid models that combine the strengths of both CNNs and Transformers have shown promising results, such as SwinURNet [23] and hybrid deep neural networks [24]. However, as the input sequence extends or the network becomes deeper, the computational complexity of the Transformer increases quadratically, which poses challenges in terms of modeling efficiency and resource requirements [25].

Mamba, a recently proposed sequence modeling architecture, addresses the computational inefficiency of Transformers in long-sequence tasks through a selective state-space model (SSM) with linear time complexity [26]. It has proven to be very promising for various image analysis tasks, including medical image classification [27], [28] and remote sensing applications [29], [30]. However, directly replacing Transformers with Mamba may not yield satisfactory results for the complex task of remote sensing semantic segmentation. As shown in Fig. 1, this task presents several challenges: 1) large variations in object scale and uneven spatial distribution of target objects often lead to missed detections or inaccurate boundaries, especially for small objects; 2) high spectral similarity among different land cover types, such as low vegetation versus trees, may lead to class confusion; and 3) significant intraclass variation in the appearance of the same class increases the difficulty of consistent recognition. These challenges require not only powerful multiscale feature extraction to handle varying object sizes but also effective modeling of LRDs (e.g., the spatial coherence of large buildings spanning distant regions) and short-range dependencies (e.g., precise localization of small objects such as cars along roadways) to improve semantic discrimination.

Therefore, we propose a multiscale convolution and Mamba fusion model (MF-Mamba). MF-Mamba adopts the classic encoder–decoder architecture, combining a CNN encoder with a Mamba-based decoder. On the one hand, multikernel atrous convolutions are employed to extract multiscale features, enabling accurate recognition of objects with varying sizes. Additionally, hierarchical features from different encoder stages are fused to preserve fine-grained spatial details. On the other hand, to address the issue of spectral similarity and class confusion among different land cover types, we design a channel–spatial attention and dense multiscale feature fusion (CSDF) module, which jointly enhances features across channel, spatial, and scale dimensions. In addition, MF-Mamba combines the local feature extraction strengths of CNNs with Mamba's ability to model LRDs, allowing the model to capture global and local contextual information. In conclusion, the key contributions of our work are as follows.

1) An MF-Mamba are proposed for remote sensing image segmentation. By integrating a CNN-based encoder with a Mamba-based decoder, MF-Mamba effectively combines local feature extraction with long-range dependency modeling. A multiscale lateral connection is further introduced to unify features from different encoder levels, enhancing the decoder's ability to recognize fine-grained targets.

2) A Mamba-based global–local state-space (GLSS) module is designed at the decoder to capture and integrate global and local information efficiently. To better model the complex spatial distributions in remote sensing scenes, an eight-direction selective scanning mechanism is introduced, enhancing the ability to model global context in multiple directions.

3) A CSDF module is developed to extract and integrate rich features from multiple perspectives. By combining multikernel atrous convolutions with channel and spatial attention mechanisms, the CSDF module enhances feature representation and improves the network's ability to distinguish confusing land cover classes.

The structure of this article is as follows. Section II reviews the related works on remote sensing image segmentation and SSMs. Section III details the architecture of MF-Mamba. Section IV describes semantic segmentation datasets and the corresponding experimental results of MF-Mamba. Section V presents ablation experiments, generalization capability, model complexity analysis, as well as a discussion on limitations and future research works. Section VI concludes the study.

II. Related Works

A. Remote Sensing Image Segmentation

Currently, the mainstream models for remote sensing image segmentation are mainly classified into two classes: CNN and Transformer. CNN extracts image features through stacked convolution operations, while the Transformer models long-distance dependencies through self-attention mechanisms. These two types of models have inspired the design of many deep model architectures, including CNN-based models, Transformer-based models, and CNN–Transformer hybrid models.

As the basis for the application of CNN-based models to remote sensing, FCN was the first to achieve end-to-end semantic segmentation. U-Net established a widely adopted symmetric encoder–decoder structure, in which the encoder progressively downsamples to capture semantic features, while the decoder reconstructs spatial resolution by

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fusing semantic and location information. Remote sensing images often exhibit significant spatial scale variation, high intraclass diversity, and low interclass contrast [31]. To effectively distinguish such complex patterns, integrating multiscale information and enhancing spatial–spectral feature representation becomes crucial [32], [33], [34]. As a result, many CNN-based models incorporate multiscale modules or attention mechanisms to improve their feature extraction capabilities. Attention-enhanced CNN models can effectively emphasize critical regions within an image, thereby refining feature representations. For example, Li et al. [35] introduced the attentive bilateral contextual network (ABCNet), which constructs a context path through a linear attention mechanism and an attention enhancement module to fully capture global contextual information. Ren et al. [36] proposed the DA-CapsUNet by introducing both channel and spatial attention modules to emphasize spatial regions most relevant to object extraction. DeepLabv3+ incorporates an atrous spatial pyramid pooling (ASPP) module for multiscale feature extraction, effectively improving segmentation accuracy while reducing computational overhead through atrous and depthwise separable convolutions. Similarly, MANet [37] leverages kernel attention and self-attention to progressively aggregate contextual information across different levels, combining densely dilated convolutions with various dilation rates to identify objects across various scales and complex shapes.

While CNN-based models are effective in learning hierarchical and multiscale local features, their limited receptive field constrains their ability to model global dependencies. To address this, Transformer-based models have been increasingly adopted in semantic segmentation tasks. Transformers use multihead self-attention to capture global contextual relationships across the entire image. For example, Xu et al. [38] proposed an efficient Transformer network based on the Swin Transformer backbone network, incorporating an MLP head to extract global information from input features. Xie et al. [39] proposed the SegFormer, which combines Transformers with a multilayer perceptron to aggregate information across multiple layers, thereby integrating both local and global attention to extract features.

Due to their strong capacity for modeling long-range interactions [40], [41], [42], Transformer-based models have shown potential in remote sensing segmentation. However, local features remain equally important, and the quadratic computational complexity of self-attention limits the scalability of pure Transformer architectures. As a result, hybrid CNN–Transformer models have emerged to combine the local modeling strength of CNNs with the global context modeling capabilities of Transformers. For example, the bilateral awareness network (BANet) [43] consists of two paths: a dependency path that captures long-range relationships using Transformers and a texture path that captures fine-grained details of the image using convolutional operations. CMTFNet [44] utilizes a hybrid architecture that fuses CNN-based local feature extraction with multiscale global context from Transformers. UNetFormer [45] employs a CNN encoder and a Transformer decoder to achieve full integration of local and global representations. Although Transformer-based and CNN–Transformer hybrid models have made significant progress in remote sensing semantic segmentation tasks, their high complexity and computational requirements still present challenges during model training and inference.

B. State-Space Models

The SSM, originally rooted in time series analysis, has gained widespread application across diverse domains, including control theory, natural language processing, and image processing. By leveraging continuous state spaces, SSM excels at modeling intricate temporal dynamics. A significant advancement in its framework is the introduction of the HiPPO initialization method [46], which substantially enhances its ability to capture LRDs, thereby facilitating its integration with deep learning. In recent years, SSMs have attracted increasing attention for their powerful capabilities in long-sequence modeling. For example, the S4 [47] model effectively simulates LRDs through diagonal parameterization, addressing computational bottlenecks. Building on S4, further developments have led to models such as S4D [48] and S5 [49], as well as the recently prominent Mamba model.

Mathematically, SSM is a sequence-to-sequence model known for its consistent time dynamics. Due to its linear complexity, SSM can map a 1-D input sequence $x(t) \in \mathbb{R}^L$ into an output sequence $y(t) \in \mathbb{R}^L$ via a hidden state $h(t) \in \mathbb{R}^N$, described by a linear ordinary differential equation (ODE)

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $h'(t)$ represents the time derivative of $h(t)$, and the state transition matrices are $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times L}$, and $\mathbf{C} \in \mathbb{R}^{L\times N}$, with N representing the size of the state. Mamba is a discretized form of this continuous system, discretizing the continuous dynamics via the zero-order hold method

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B} \end{aligned} \tag{2}$$

where Δ represents the time scale parameter, used to convert the continuous parameters A and B into the discrete counterparts $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. After discretization, the discrete SSM is obtained

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

Finally, the output can be expressed in the form of a convolution, as shown in the following equation:

$$\begin{aligned} \bar{\mathbf{K}} &= \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}\right) \\ y &= x^* \bar{\mathbf{K}} \end{aligned} \tag{4}$$

where * indicates the convolution operation and L represents the length of the sequence x.

Mamba is a novel SSM that uses a selective scanning mechanism to convert time-invariant parameters into time-varying parameters and integrates them into the SSM to enhance its flexibility. For example, static model weights $\mathbf{B}$, $\mathbf{C}$, and Δ are replaced with parameters that can adapt based on both the current and historical context of the input sequence. Therefore, Mamba can more effectively capture complex temporal dynamics.

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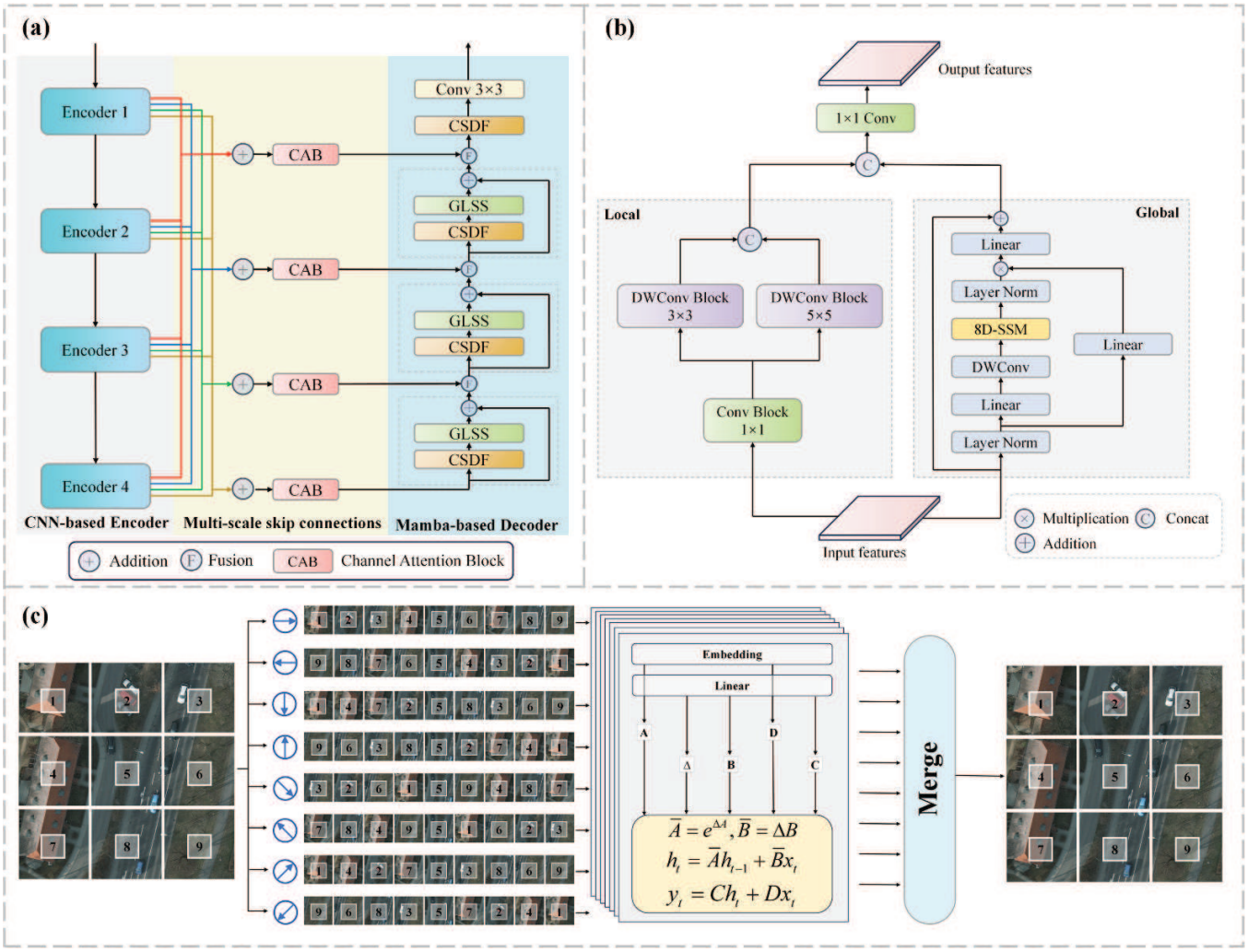


Fig. 2. Overall network structure of MF-Mamba. (a) MF-Mamba. (b) GLSS. (c) 8D-SSM.

Due to Mamba's ability to model LRDs with linear computational complexity, it has found widespread applications in the visual domain. For instance, VMamba [50] transfers Mamba's selective SSM design from natural language processing to image understanding by introducing the SS2D module, which bridges sequential 1-D processing and nonsequential 2-D feature traversal. U-Mamba incorporates Mamba blocks into a U-Net-like network structure, combining CNN's local patterns with Mamba's global context to achieve high-precision medical image segmentation. For remote sensing data, RSMamba [51] employs Mamba for image classification tasks and introduces a dynamic multipath activation module, VSS, to enhance its ability to model noncausal data. RS3Mamba [52] proposes a dual-branch network in which a convolution-based main branch is provided with additional global information by an auxiliary branch utilizing VSS blocks. Samba [53] constructs an encoder using Mamba to efficiently extract multilevel semantic features for high-resolution remote sensing segmentation.

However, most of these methods simply use Mamba as a substitute for the Transformer to capture spatial context from the input images, without considering the importance of multiscale information and attention mechanisms in remote sensing image semantic segmentation. Moreover, these models typically integrate Mamba in the encoder and ignore its potential in the decoder. More importantly, since remote sensing imagery is acquired from a top–down perspective via airborne or satellite platforms, the spatial patterns of the manifold target objects can appear in any orientation. Therefore, Mamba requires a more flexible scanning mechanism.

III. Methodology

A. Overall Structure

The overall network structure of MF-Mamba is shown in Fig. 2(a). MF-Mamba consists of three parts: a CNN-based encoder, multiscale lateral connections, and a Mamba-based decoder. Throughout training, the CNN-based encoder extracts hierarchical features of different resolutions from the input image through convolution and downsampling. Multiscale lateral connections fuse features from different encoder layers, passing them to the decoder to fully mine and utilize the details of objects under different scale features. The Mamba-based decoder includes two main components: the GLSS module and the CSDF module. During the decoding process, the GLSS module uses Mamba for global relationship modeling, while

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convolution operations are employed to extract local features. The CSDF module further enhances feature representation by applying channel and spatial attention, as well as densely connected multikernel convolutions, to aggregate channel, spatial, and multiscale information. To effectively integrate semantic information from the decoder with spatial data from the encoder, the fusion module performs weighted fusion of the features from both components using learnable parameters. Finally, at the end of the decoder, the feature map is resized to match the input image dimensions using convolution and interpolation, producing the final prediction outcome.

B. GLSS Module

Transformer is good at extracting global context using a self-attention mechanism, but it suffers from computational inefficiency due to its quadratic complexity. Mamba is an SSM that is not only good at capturing global information but also has linear complexity, making it a promising alternative to Transformer. However, the scanning operation in Mamba processes data sequentially along a specific direction, which may not be sufficient to fully model the spatial dependencies of objects in various directions in remote sensing images. Furthermore, in the remote sensing imagery segmentation task, local feature information is equally important as global context. However, Mamba often ignores local semantic information when processing high-resolution features. To address these challenges, we developed the GLSS module, illustrated in Fig. 2(b), which features a dual-branch architecture. One branch uses Mamba to replace Transformer and introduces an eight-directional scanning mechanism to extract global information in multiple directions, while the other employs parallel multicore deep convolution operations to capture local information. Finally, the two are combined to achieve a more comprehensive feature representation.

Specifically, for the input feature $\mathbf{X}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, we process it through two distinct branches: the global branch and the local branch. The global branch is constructed based on the VSSBlock [50], which mainly includes a normalization layer (LN), a linear layer (Linear), a depthwise convolution (dwConv), and an eight-direction selective scanning space state sequence model (8D-SSM). Based on the input feature $\mathbf{X}_{\text{in}}$, it generates an output $\mathbf{X}_{\text{Global}}$ with the same size as the input

$$\begin{aligned}\mathbf{X}_1 &= \text{LN}\left(8D-\text{SSM}\left(\text{dwConv}\left(\text{Linear}\left(\text{LN}\left(\mathbf{X}_{\text{in}}\right)\right)\right)\right)\right)\\ \mathbf{X}_{\text{Global}} &= \text{Linear}\left(\mathbf{X}_1 \otimes \left(\text{Linear}\left(\text{LN}\left(\mathbf{X}_{\text{in}}\right)\right)\right)\right)+\mathbf{X}_{\text{in}}\end{aligned} \tag{5}$$

where $\otimes$ denotes the Hadamard product. To enable Mamba to capture spatial features of remote sensing images from multiple directions and model global information, we constructed an eight-directional scanning SSM (8D-SSM) based on OSSM [25], as illustrated in Fig. 2(c). The 8D-SSM initially unrolls the input patches into sequential representations following eight different traversal paths, including horizontal, vertical, diagonal, and anti-diagonal directions, in both forward and backward orientations. Each patch sequence is then processed in parallel using separate Mamba blocks, followed by reshaping and merging the structures to generate the output map. The 8D-SSM effectively scans the input image from multiple directions, enabling each pixel to acquire information from all other pixels across various directions, capturing complex spatial relationships and extracting global information.

The local branch includes a 1×1 convolution block and two parallel depthwise separable convolutions of 3×3 and 5×5. Given the input feature $\mathbf{X}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, the results of these convolutions at different scales are then concatenated to obtain the local feature $\mathbf{X}_{\text{Local}}$

$$\begin{aligned}\mathbf{X}_{3\times 3} &= \text{dwConv}_{3\times 3}\left(\text{Conv}_{1\times 1}\left(\mathbf{X}_{\text{in}}\right)\right)\\ \mathbf{X}_{5\times 5} &= \text{dwConv}_{5\times 5}\left(\text{Conv}_{1\times 1}\left(\mathbf{X}_{\text{in}}\right)\right)\\ \mathbf{X}_{\text{Local}} &= \text{Concat}\left(\mathbf{X}_{3\times 3}, \mathbf{X}_{5\times 5}\right)\end{aligned} \tag{6}$$

where $\text{Conv}_{1\times 1}(\cdot)$ denotes a 1×1 convolution module, which includes a Conv2D layer, a BatchNorm2D normalization layer, and a ReLU activation function. $\text{dwConv}_{\text{k}\times\text{k}}(\cdot)$ denotes a depthwise convolution block with a kernel size of k. Using depthwise separable convolutions helps maintain the expressive power of convolutional features while reducing the number of parameters. To enhance the network's ability to perceive fine-grained details, multiple kernel sizes are used to capture local features at different spatial scales. Finally, the global and local features are concatenated along the channel dimension. The module's final output is obtained through a 1×1 convolution, which restores the channel count to match the input channels for the residual connection.

C. CSDF Module

To effectively capture multiscale features and spatial channel information, we propose the CSDF module that combines multikernel atrous convolution with attention mechanisms, as illustrated in Fig. 3. This module consists of three branches: the channel attention branch, the dense multiscale connection branch, and the spatial attention branch. Given an input feature map, each branch processes it independently to extract different types of informative features. The outputs of all three branches are subsequently fused via element-wise summation to generate the final enhanced feature representation.

The channel attention branch is designed to enhance the expression capability of the channel information in the feature maps. Initially, the input feature $\mathbf{X}_{\text{in}}$ undergoes global average pooling to produce the channel attention map $\mathbf{X}_c$. This map is then processed through two consecutive 1×1 convolution layers followed by ReLU activation functions. Subsequently, it is processed through a sigmoid function to obtain the channel attention weights. These weights are then applied element-wise to the original input feature $\mathbf{X}_{\text{in}}$, yielding the enhanced feature map $\mathbf{X}_{\text{channel}}$. This process can be represented as follows:

$$\begin{aligned}\mathbf{X}_c &= \text{AvgPool}\left(\mathbf{X}_{\text{in}}\right)\\ \mathbf{X}_{\text{channel}} &= \mathbf{X} \otimes \text{Sigmoid}\left(\text{Conv}_{1\times 1}\left(\text{ReLu}\left(\text{Conv}_{1\times 1}\left(\mathbf{X}_c\right)\right)\right)\right).\end{aligned} \tag{7}$$

In this way, the channel attention branch can adaptively adjust the significance of each channel, thereby enhancing the model's ability to concentrate on essential information.

The dense multiscale connection branch aims to capture the features of objects across varying scales in remote sensing images. This branch densely connects a set of depthwise separable convolutional layers with distinct dilation rates, generating multiscale features that effectively expand the

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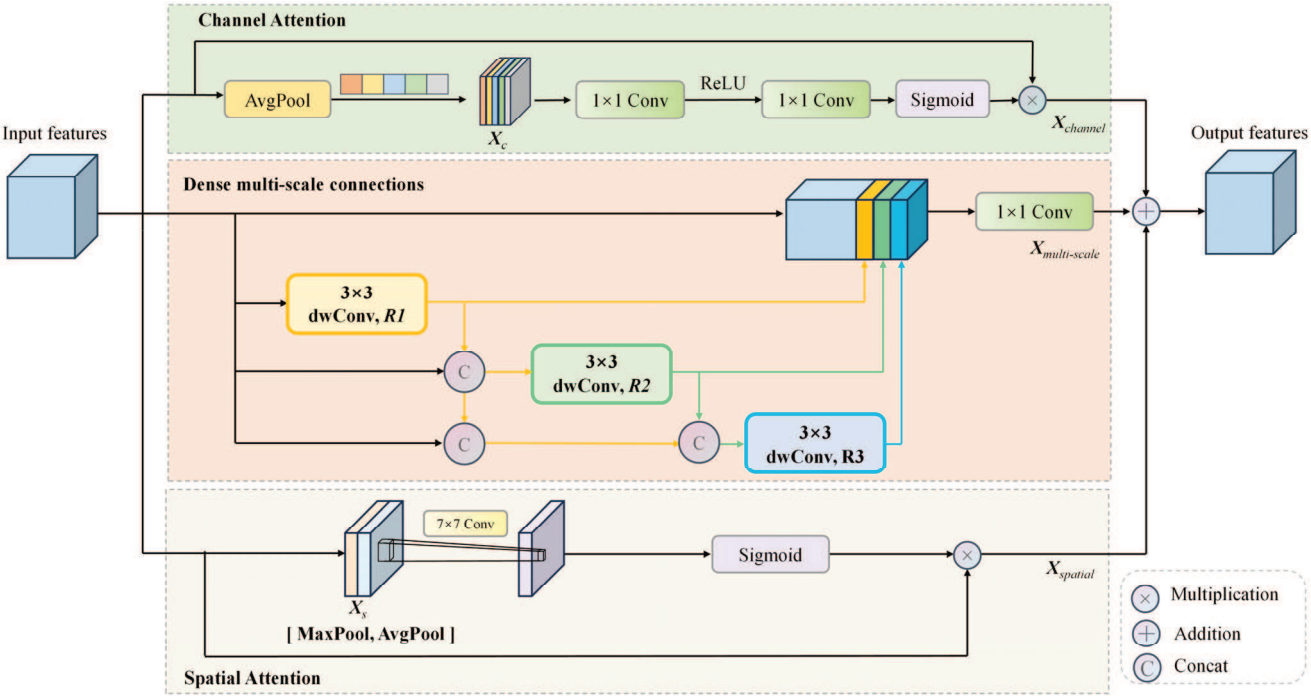


Fig. 3. Illustration of the CSDF module.

receptive field without significantly increasing model parameters. Specifically, the multiscale branch consists of three depthwise separable convolution layers with 3×3 kernels and dilation rates $R_1 - R_3$ of 6, 12, and 18, respectively. This design accommodates the considerable variations in object sizes present in remote sensing imagery. By incorporating multiple dilation rates, the model effectively captures features across different scales, enabling comprehensive contextual modeling for more accurate segmentation results. The output from each convolutional layer is concatenated with both the original input feature map and the outputs from previous layers, which are passed to the next convolutional layer. To limit computational cost from excessive channel accumulation, each depthwise separable convolution outputs a fixed number of channels, maintaining a balance between representational capacity and efficiency. Finally, the outputs from all convolution layers are concatenated and passed through a 1×1 convolution to restore the original channel dimensions, resulting in the multiscale enhanced feature map $\mathbf{X}_{\text{multi-scale}}$.

To capture spatial information, the spatial attention branch first adopts average pooling and max pooling to generate feature maps $\mathbf{X}_s$. These feature maps are concatenated in the channel dimension and passed through a 7×7 convolution layer. The output is processed through sigmoid to produce the spatial attention weights. The obtained weights are element-wise multiplied with the original input $\mathbf{X}_{\text{in}}$, resulting in the spatial attention-enhanced feature map $\mathbf{X}_{\text{spatial}}$

$$\begin{aligned} \mathbf{X}_s &= \text{Concat}\left(\text{MaxPool}\left(\mathbf{X}_{\text{in}}\right), \text{AvgPool}\left(\mathbf{X}_{\text{in}}\right)\right) \\ \mathbf{X}_{\text{spatial}} &= \text{Sigmoid}\left(\text{Conv}_{7\times7}\left(\mathbf{X}_s\right)\right) \otimes \mathbf{X}_{\text{in}}. \end{aligned} \tag{8}$$

Ultimately, the features from all three branches are combined by summation to produce the final output $\mathbf{X}_{\text{out}}$.

D. Multiscale Lateral Connections

In traditional encoder–decoder networks, lateral connections transfer feature maps from the encoder to the decoder, leveraging the spatial detail preserved in shallow convolutional layers to enhance the semantic representation of deeper layers. This design improves the network's overall capacity for semantic extraction. However, due to the significant variation in object sizes in high-resolution remote sensing images, models like U-Net, which simply connect corresponding encoder and decoder layers, struggle to fully exploit the multiscale features present in the feature maps. To overcome this limitation, we introduce multiscale lateral connections that enable a more comprehensive capturing and integration of multiscale information between the encoder and the decoder. This design enhances the network's ability to handle remote sensing scenes with diverse spatial scales.

The redesigned lateral connections modify the connectivity between the encoder and the decoder, enabling decoder layers to access feature information not only from the corresponding encoder layer but also from other encoder layers, as illustrated in Fig. 4. Specifically, each decoder layer receives two types of input: 1) feature information aggregated from multiple encoder layers via multiscale lateral connections and 2) the upsampled output of the previous decoder layer through transposed convolution. For the multiscale encoder features, 1×1 convolutions are first applied to align the channel dimensions with those of the corresponding decoder layer. Features from the same level are directly used, while those from different encoder layers are resized using bilinear interpolation to match the spatial resolution of the decoder.

The aligned feature maps are then aggregated, and a channel attention module is applied to refine the feature response before forwarding to the decoder. By summing these two parts,

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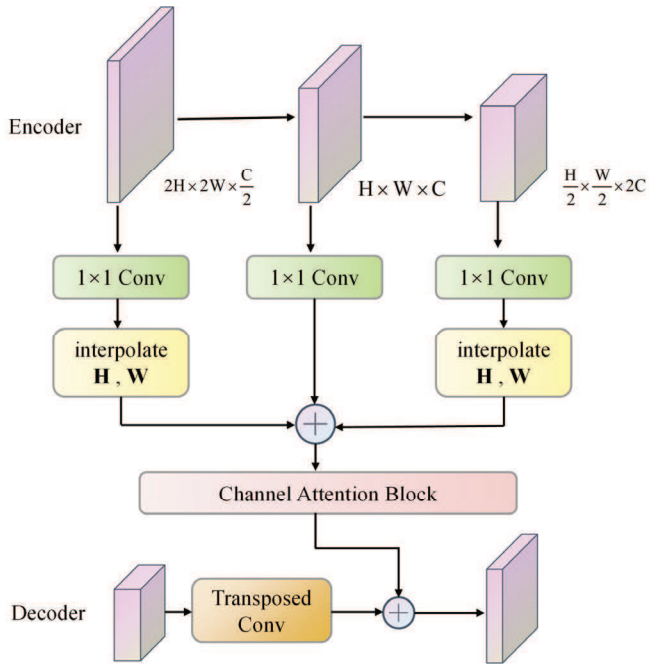


Fig. 4. Illustration of the multiscale lateral connections.

the input for the corresponding decoder layer is obtained. This improvement effectively utilizes the multiscale feature information of different layers of the encoder.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Metrics

To assess the performance of the proposed model, we conducted experiments on three publicly available high-resolution remote sensing datasets: ISPRS Vaihingen, ISPRS Potsdam, and Wuhan Dense Labeling Dataset (WHDLD) [54].

1) ISPRS Vaihingen: The Vaihingen dataset comprises 33 orthophotos, each with an average resolution of 2494 × 2064 pixels and a spatial resolution of 9 cm. Each image includes three bands: red, green, and near infrared, and is annotated with six land cover classes: impervious surface, building, low vegetation, tree, car, and clutter/background. Following existing studies [35], [44], the number of images used for training, validation, and testing is 15, 1, and 17, respectively.

2) ISPRS Potsdam: It comprises 38 high-resolution images, each measuring 6000 × 6000 pixels with a spatial resolution of 5 cm, and annotated with the same classes as the Vaihingen dataset. In our experiments, we used the red, green, and blue bands contained in the image. Similarly, the number of images used for training, validation, and testing is 22, 1, and 14, respectively.

3) WHDLD Dataset: It comprises 4940 RGB images, each measuring 256 × 256 pixels with a spatial resolution of 2 m. Every pixel in these images is manually labeled into six classes: bare soil, pavement, building, vegetation, road, and water. We randomly split the dataset into training, validation, and testing sets in a ratio of 0.7:0.1:0.2. The final number of training images, validation images, and test images was 3458, 494, and 988, respectively. To evaluate the model's performance, we use three common metrics: $F1$ score ($F1$), mean intersection over union (mIoU), and overall accuracy (OA) [56]. These metrics can be derived from the cumulative confusion matrix, with the specific formulas as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{K} \mathrm{TP}_k}{\sum_{k=1}^{K} \mathrm{TP}_k + \mathrm{FP}_k + \mathrm{TN}_k + \mathrm{FN}_k} \tag{9}$$

$$\mathrm{mIoU} = \frac{1}{K} \sum_{k=1}^{K} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{10}$$

$$\mathrm{precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{11}$$

$$\mathrm{recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \tag{12}$$

$$F1_k = 2 \times \frac{\mathrm{precision}_k \times \mathrm{recall}_k}{\mathrm{precision}_k + \mathrm{recall}_k} \tag{13}$$

where K represents the number of classes and TP_k, FP_k, TN_k, and FN_k denote the true positives, false positives, true negatives, and false negatives for class k, respectively.

B. Experimental Setting

All experiments were trained and tested with PyTorch on a single NVIDIA RTX 3090 GPU. The model was optimized with the AdamW optimizer, starting with a learning rate of 0.0005 and a weight decay coefficient of $2e^{-4}$. Training was conducted for up to 50 epochs using cross entropy as the loss function. HRNet18 was adopted as the encoder of our model. To balance model performance and computational efficiency, a sliding window mechanism was used. For the Vaihingen and Potsdam datasets, training images were divided into 512 × 512 patches with a stride of 256 pixels. For the WHDLD dataset, we used the original 256 × 256 patches directly. Since the image sizes vary across datasets, we adjusted the batch size accordingly: 8 for Vaihingen and Potsdam, and 32 for WHDLD. To mitigate overfitting, we used several data augmentation methods, such as random vertical flipping, random horizontal flipping, and random rotation.

C. Results and Comparison With State-of-the-Art Methods

We conducted a comprehensive evaluation of the proposed MF-Mamba and compared it to several current mainstream deep learning models for semantic segmentation. These comparative models include CNN-based architectures such as Deeplabv3+ [14], DDCMNet [55], MANet [37], and ABCNet [35]; Transformer-based hybrid architectures such as SegFormer [39], BANet [43], UnetFormer [45], and CMTFNet [44]; and Mamba-based architectures such as RS3Mamba [52].

1) ISPRS Vaihingen Dataset: Table I presents the experimental results of different methods applied on the ISPRS Vaihingen dataset. To ensure a fair and accurate evaluation, we exclude the clutter/background class due to its negligible proportion in the dataset and its vague semantic definition, which offers limited value in assessing segmentation performance. The results demonstrate that MF-Mamba achieves the highest $F1$ score, mIoU, and OA. Compared with other methods, MF-Mamba improves the MeanF1 and mIoU metrics by more than 0.68% and 1.02%, respectively. MF-Mamba excels particularly

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TABLE I
QUANTITATIVE RESULTS OF EXISTING STATE-OF-THE-ART MODELS ON THE ISPRS VAIHINGEN

Method	F1 (%)					Evaluation metrics (%)		
	Imp.suf	Building	Lowveg	Tree	Car	MeanF1	mIoU	OA
Deeplabv3+ [14]	89.83	93.54	79.87	86.22	79.07	85.71	75.41	87.67
ABCNet [35]	89.68	93.18	78.85	85.53	79.02	85.25	74.73	87.15
SegFormer [39]	89.21	92.51	78.77	85.92	78.51	84.98	74.30	86.97
UnetFormer [45]	89.84	93.40	79.55	86.03	80.28	85.82	75.55	87.51
CMTFNet [44]	89.88	94.04	80.03	**86.32**	79.86	86.03	75.90	87.88
MANet [37]	89.94	93.58	79.54	85.97	80.00	85.81	75.55	87.59
DDCMNet [55]	89.51	93.41	79.99	86.03	79.42	85.67	75.33	87.49
BANet [43]	89.36	93.19	78.94	85.81	77.83	85.03	74.41	87.17
RS3Mamba [52]	89.74	93.69	79.84	86.25	78.51	85.61	75.28	87.66
MF-Mamba	**90.65**	**94.36**	**80.87**	86.16	**81.53**	**86.71**	**76.92**	**88.30**

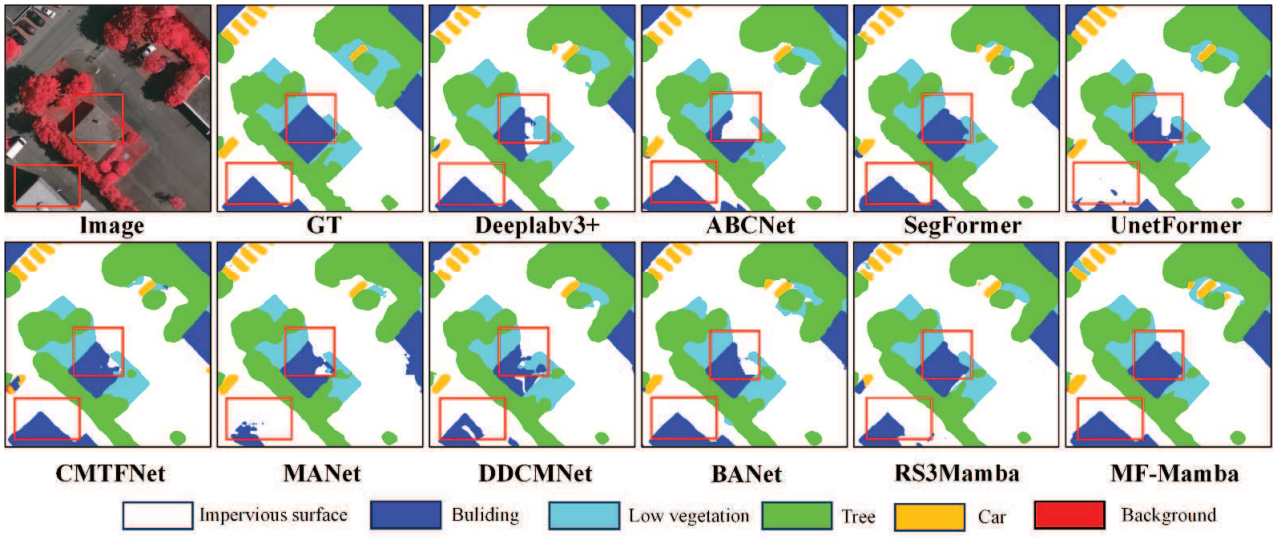


Fig. 5. Qualitative comparison results on the ISPRS Vaihingen.

in the thematic classes of impervious surfaces, building, low vegetation, and car, achieving the highest $F1$ scores across these classes. Notably, the "car" class is especially challenging due to its small size, uneven distribution, and limited training samples. However, MF-Mamba's integration of global–local fusion, multikernel convolutions, and multiscale lateral connections significantly enhances the network's ability to capture small-scale objects. As a result, MF-Mamba achieves an $F1$ score for the "car" class that is over 1.25% higher than all baseline methods.

Fig. 5 shows the visual results for each method. ABCNet, while utilizing a dual-branch attention mechanism that captures spatial detail effectively, tends to misclassify buildings as impervious surfaces due to inadequate spectral feature utilization. UnetFormer extracts global and local contextual information through a Transformer-based architecture and performs well in capturing fine-grained spatial details but is still limited in distinguishing between "building" and "impervious surface." In contrast, MF-Mamba not only captures global and spatial context but also employs a channel attention mechanism to model spectral information effectively, leading to improved discrimination between spectrally similar classes like "building" and "impervious surface."

2) ISPRS Potsdam Dataset: Table II presents the experimental results applied on the ISPRS Potsdam, which features higher resolution and wider coverage than the ISPRS Vaihingen dataset. As with ISPRS Vaihingen, the clutter/background class is excluded from evaluation. All models achieve Mean $F1$ scores above 89%, but MF-Mamba surpasses them all in Mean $F1$, mIoU, and OA. Specifically, MF-Mamba improves Mean $F1$ and mIoU by 0.36% and 0.56%, respectively, over the second-best method. SegFormer proposed a hierarchical Transformer encoder that can extract global and local information as well as multiscale features, but its relatively single input sequence direction limits the applicability of the model to remote sensing images. In contrast, MF-Mamba employs an eight-directional scanning mechanism via its SSM, enabling it to capture edge features and object arrangements more comprehensively. It achieves the highest $F1$ scores across all classes. Fig. 6 presents the visual results. While other models frequently misclassify shadowed cars as impervious surfaces, MF-Mamba accurately segments them, demonstrating its robustness in complex scenes.

3) WHDLD Dataset: Table III presents the experimental results applied on the WHDLD, where MF-Mamba again

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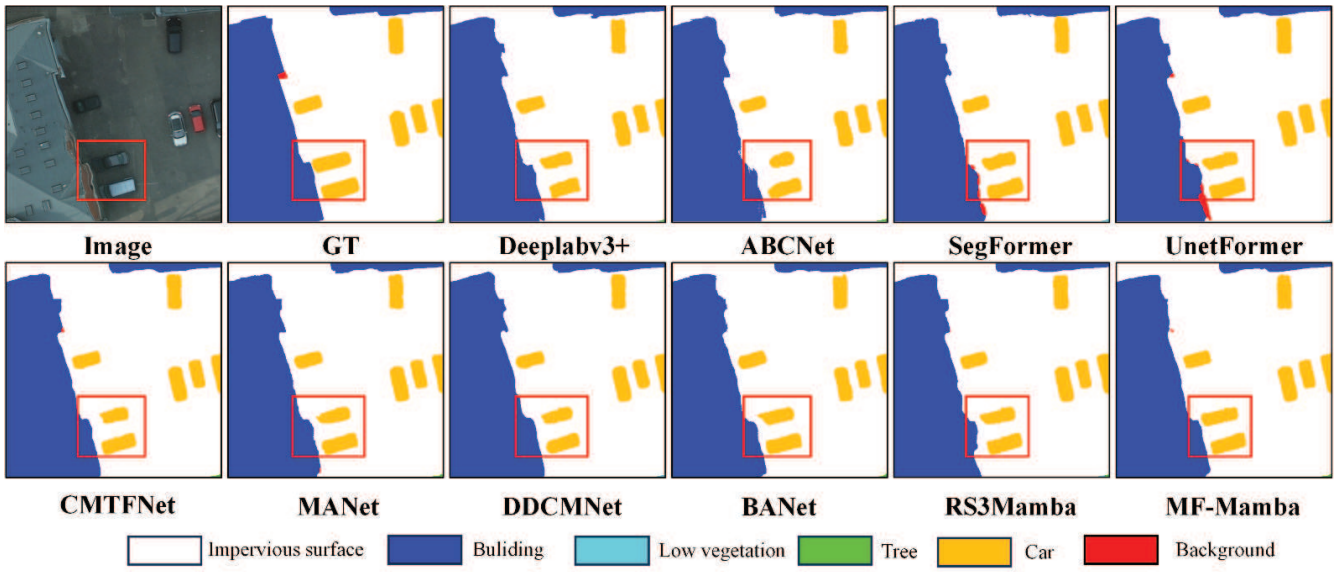


Fig. 6. Qualitative comparison results on the ISPRS Potsdam.

TABLE II
QUANTITATIVE RESULTS OF EXISTING STATE-OF-THE-ART MODELS ON THE ISPRS POTSDAM

Method	F1 (%)					Evaluation metrics (%)		
	Imp.suf	Building	Lowveg	Tree	Car	MeanF1	mIoU	OA
Deeplabv3+ [14]	92.41	96.57	85.49	85.73	91.39	90.32	82.61	90.24
ABCNet [35]	92.18	96.24	85.27	85.41	89.88	89.80	81.75	89.95
SegFormer [39]	92.00	95.67	85.26	85.28	90.17	89.68	81.53	89.72
UnetFormer [45]	92.26	96.32	85.37	84.77	91.76	90.10	82.27	89.76
CMTFNet [44]	92.77	96.58	85.62	85.19	91.56	90.34	82.68	90.28
MANet [37]	92.33	95.98	85.36	85.73	91.36	90.15	82.32	89.93
DDCMNet [55]	92.45	96.26	85.58	85.28	91.17	90.15	82.33	90.05
BANet [43]	92.03	95.78	85.49	85.44	90.93	89.93	81.95	89.86
RS3Mamba [52]	92.17	96.15	85.59	85.64	91.00	90.11	82.25	90.13
MF-Mamba	**92.88**	**96.71**	**85.94**	**86.12**	**91.83**	**90.70**	**83.24**	**90.60**

TABLE III
QUANTITATIVE RESULTS OF EXISTING STATE-OF-THE-ART MODELS ON THE WHDLD

Method	F1 (%)						Evaluation metrics (%)		
	building	road	pavement	vegetation	bare soil	water	MeanF1	mIoU	OA
Deeplabv3+ [14]	74.21	75.71	61.22	89.81	56.50	96.99	75.74	63.17	84.36
ABCNet [35]	72.88	73.81	59.50	89.48	56.31	96.76	74.79	62.01	83.83
SegFormer [39]	73.63	74.49	60.67	89.41	56.15	96.66	75.17	62.43	83.96
UnetFormer [45]	74.19	74.46	60.13	89.80	56.02	97.00	75.27	62.64	84.26
CMTFNet [44]	75.14	76.34	62.23	90.04	57.38	97.11	76.37	63.93	84.82
MANet [37]	75.10	75.58	61.67	90.22	56.85	97.17	76.10	63.64	84.80
DDCMNet [55]	72.83	75.33	59.62	89.51	56.95	97.01	75.21	62.53	83.97
BANet [43]	74.14	74.57	61.17	89.77	56.12	97.05	75.47	62.86	84.34
RS3Mamba [52]	74.04	73.98	59.93	89.98	55.32	96.98	75.03	62.41	84.26
MF-Mamba	**76.32**	**77.10**	**62.96**	**90.41**	**58.43**	**97.22**	**77.07**	**64.79**	**85.26**

achieves the best performance across Mean $F1$, mIoU, and OA. In this dataset, classes such as building, road, pavement, and bare soil exhibit highly similar texture and color characteristics, making them difficult to distinguish. MANet, though effective in segmenting distinct classes like vegetation and water, performs poorly on these more ambiguous classes due to limited consideration of spectral and spatial context. RS3Mamba attempts to enhance CNN-based encoders with global information from a Mamba-based auxiliary branch. However, its lack of joint modeling of spectral–spatial features

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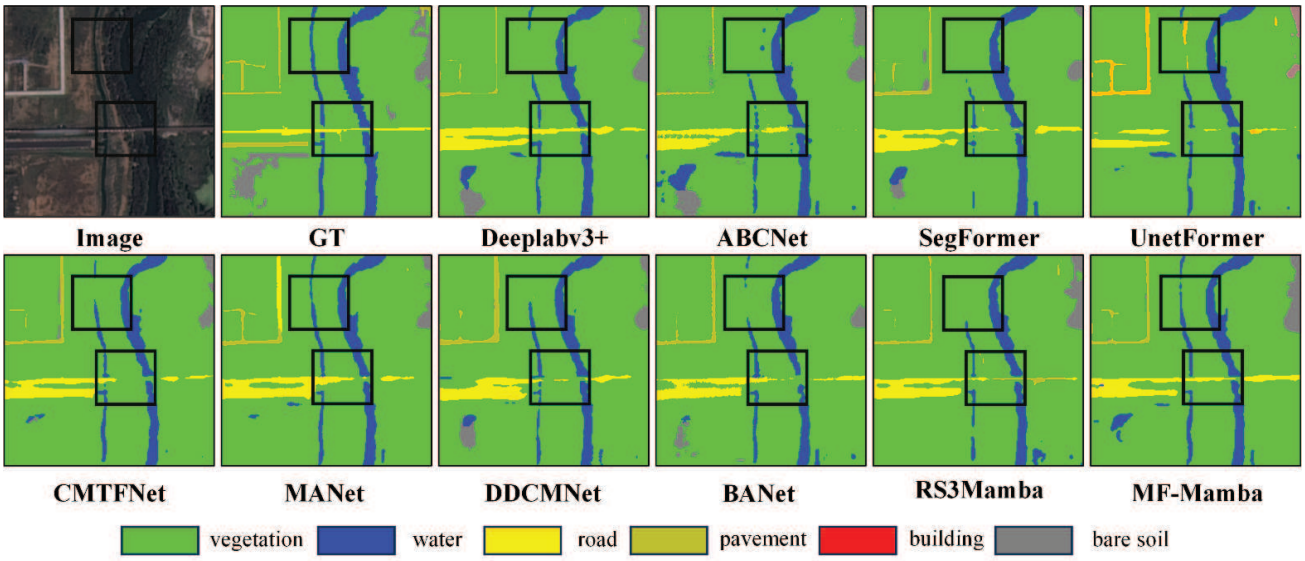


Fig. 7. Qualitative comparison results on the WHDLD dataset.

TABLE IV
ABLATION STUDY RESULTS OF MF-MAMBA ON THE ISPRS VAIHINGEN DATASET

Method	F1 (%)					Evaluation metrics (%)		
	Imp.suf	Building	Lowveg	Tree	Car	MeanF1	mIoU	OA
FCN-8S [10]	86.51	89.52	75.73	85.01	59.12	79.19	66.83	84.35
baseline	89.35	93.15	79.04	85.61	77.42	84.91	74.25	87.12
baseline+GLSS	90.16	93.64	79.99	86.24	80.49	86.10	75.98	87.81
baseline+GLSS+CSDF	90.25	94.09	80.19	**86.31**	81.49	86.46	76.53	88.04
MF-Mamba	**90.65**	**94.36**	**80.87**	86.16	**81.53**	**86.71**	**76.92**	**88.30**

TABLE V
ABLATION STUDY RESULTS OF MF-MAMBA ON THE ISPRS POTSDAM DATASET

Method	F1 (%)					Evaluation metrics (%)		
	Imp.suf	Building	Lowveg	Tree	Car	MeanF1	mIoU	OA
FCN-8S [10]	90.21	94.33	83.60	84.24	87.52	87.98	78.77	88.24
baseline	92.25	96.02	85.53	84.66	90.41	89.77	81.71	89.90
baseline+GLSS	92.85	**96.76**	86.12	85.88	91.56	90.63	83.14	90.59
baseline+GLSS+CSDF	**92.95**	96.57	**86.14**	86.11	91.51	90.66	83.16	90.59
MF-Mamba	92.88	96.71	85.94	**86.12**	**91.83**	**90.70**	**83.24**	**90.60**

and multiscale context results in suboptimal performance under complex scenarios. In contrast, MF-Mamba achieves the highest $F1$ scores across all challenging classes. Notably, its accuracy for the building and bare soil classes surpasses that of other methods by more than 1.18% and 1.05%, respectively. This demonstrates MF-Mamba's ability to effectively distinguish semantically similar regions through contextual modeling of spectral and spatial cues.

Fig. 7 provides visual comparisons. In scenes where water or road elements are narrow, linear, and partially obscured by background interference, other models have difficulties to extract these in a comprehensive manner and show obvious discontinuity. In contrast, MF-Mamba delivers more complete and continuous segmentation, verifying its effectiveness in handling complex remote sensing imagery.

V. DISCUSSION

A. Ablation Experiments

1) Effects of Each Module of MF-Mamba: To evaluate the performance of each module in MF-Mamba, we conducted ablation experiments on the ISPRS Vaihingen, ISPRS Potsdam, and WHDLD. By incrementally adding different modules, we analyzed their effects on model performance to verify their effectiveness. We built a baseline model using FCN8s with an HRNet18 encoder. Subsequently, we gradually added the GLSS module, the CSDF module, and multiscale lateral connections, ultimately constructing the complete MF-Mamba model. To further demonstrate the superiority of HRNet18 as an encoder, we also compare it with the

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TABLE VI
ABLATION STUDY RESULTS OF MF-MAMBA ON THE WHDLD DATASET

Method	F1 (%)						Evaluation metrics (%)		
	building	road	pavement	vegetation	bare soil	water	MeanF1	mIoU	OA
FCN-8S [10]	68.91	69.92	54.29	88.17	52.20	95.80	71.55	58.28	81.71
baseline	73.51	73.99	59.15	89.42	55.67	96.64	74.73	61.96	83.68
baseline+GLSS	76.11	76.60	62.38	90.37	56.47	97.21	76.52	64.20	85.09
baseline+GLSS+CSDF	75.98	76.82	62.51	90.35	57.84	**97.23**	76.79	64.47	85.16
MF-Mamba	**76.32**	**77.10**	**62.96**	**90.41**	**58.43**	97.22	**77.07**	**64.79**	**85.26**

original FCN-8S [10] with the VGG16 encoder. The results are presented in Tables IV–VI.

a) Baseline: HRNet18 maintains high-resolution representations throughout the network while offering low computational complexity and strong semantic expressiveness, which is beneficial for dense prediction tasks like remote sensing image segmentation. Therefore, we built a representative baseline model based on the FCN-8S architecture and used HRNet18 as the encoder. Compared with the original FCN-8s (VGG16 encoder), the HRNet18-based baseline achieved significant improvements in Mean $F1$, mIoU, and OA on the three datasets. Notably, on the Vaihingen dataset, the $F1$ score for the "car" class improved by18.3%, validating the effectiveness of HRNet18 as a feature extractor.

b) Baseline+GLSS: The GLSS module combines Mamba's eight-directional scanning mechanism and parallel multikernel convolution to fuse global and local spatial information, thereby enhancing the model's interpretation ability in complex scenes. After integrating the GLSS module into the baseline model, we observed consistent improvements across all performance metrics on all datasets.

c) Baseline+GLSS+CSDF: To further enhance feature representation, we added the CSDF module, which leverages channel attention, multiscale context, and spatial information through densely connected atrous convolutions and attention mechanisms. The experimental results show that the inclusion of CSDF improved the average $F1$ score on the Vaihingen dataset by 0.36% and mIoU by 0.55%. The $F1$ score for the "car" class increased by 1%, reaching 81.49%. On the WHDLD dataset, the addition of CSDF also led to gains of 0.27% in both Mean $F1$ and mIoU, confirming the effectiveness of the module.

d) MF-Mamba: Finally, we introduced the multiscale lateral connection to construct the complete MF-Mamba network. Multiscale lateral connections allow the model to fuse features from various scales of the encoder during the decoding process and fully explore the details of objects under different scale features in the encoder. As shown in Table IV, the MF-Mamba model achieved the highest $F1$ scores in most classes, and the Mean $F1$, mIoU, and OA metrics also reached their highest values. Notably, the $F1$ score for the small-scale "car" class reached 81.53%, representing a 4.11% improvement over the baseline, the largest gain among all classes. Similarly, on the Potsdam dataset, MF-Mamba achieved an $F1$ score of 91.83% for cars, 1.42% higher than the baseline. This indicates that MF-Mamba can better capture and integrate multiscale feature information, significantly enhancing the segmentation capabilities for small-scale objects. In the WHDLD dataset, the "bare soil" class poses significant challenges due to its spectral similarity with roads and pavement. MF-Mamba integrates spectral and spatial information through effective context modeling to distinguish these objects with similar features, making the accuracy of the "bare soil" class 2.76% higher than the baseline model.

2) Visualizations of Network Structures: To further validate the effectiveness of MF-Mamba, we conducted a detailed analysis of its key modules by visualizing the feature activation patterns using class activation mapping (CAM). We compared the activation maps of FCN-8 s, baseline, baseline+GLSS, baseline+GLSS+CSDF, and the complete MF-Mamba, as shown in Fig. 8. The top two rows correspond to the "building" class, while the bottom two rows represent the "car" class. In the first row, adding the GLSS module significantly enhances the response intensity for the "building" class, demonstrating that the fusion of global and local spatial information improves semantic segmentation in complex scenes. MF-Mamba further strengthens these activations while effectively suppressing erroneous responses from nontarget classes (e.g., impervious surfaces), reducing false positives. In the second row, the addition of the CSDF module leads to a more complete and coherent response for small-scale buildings. This improvement is attributed to the fusion of spectral–spatial features and multiscale information, which helps mitigate issues caused by scale variation and missing detections. In the third and fourth rows, which focus on the "car" class, models without the GLSS module show weak and fragmented activations, especially along object boundaries. After introducing global context modeling and multiscale feature extraction, the activation maps become more focused and intense. These results indicate that MF-Mamba significantly improves the segmentation accuracy for small-scale objects by leveraging comprehensive feature modeling. Overall, these visualizations highlight the robustness of MF-Mamba in handling objects of varying sizes and support the effectiveness of the proposed global–local information fusion and multiscale modeling strategies.

3) Effects of Selective Scanning Mechanism in GLSS: We further analyzed the impact of different directional scanning configurations in the GLSS module. The configurations include single-direction (1-D), two-direction (2-D), four-direction (4-D), and eight-direction (8-D) selective scanning, corresponding to combinations of horizontal, vertical, diagonal, and anti-diagonal scanning paths (see Fig. 9). The experimental results, summarized in Table VII, show that the 8-D scanning configuration adopted in MF-Mamba achieves the best performance. Specifically, it improves Mean $F1$ and

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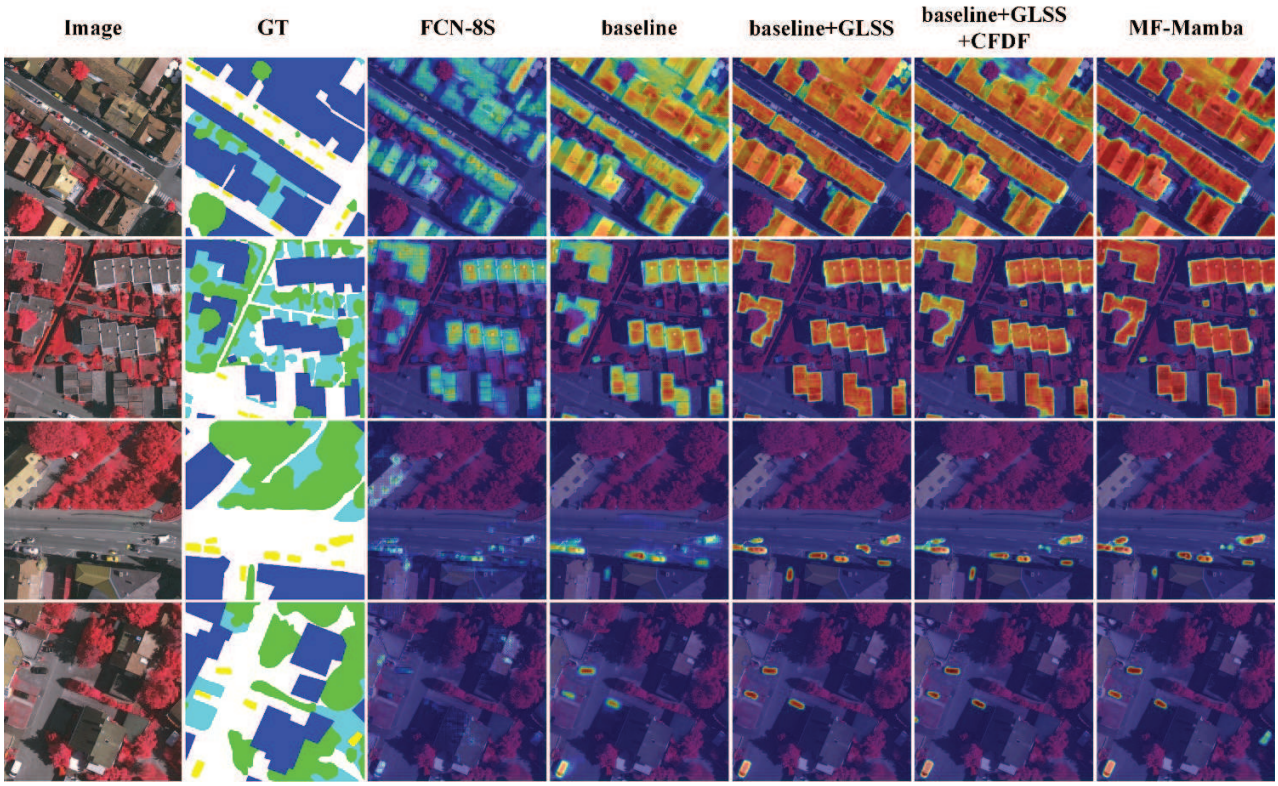


Fig. 8. Class activation maps of different network configurations for building (first and second rows) and car (third and fourth rows) classes.

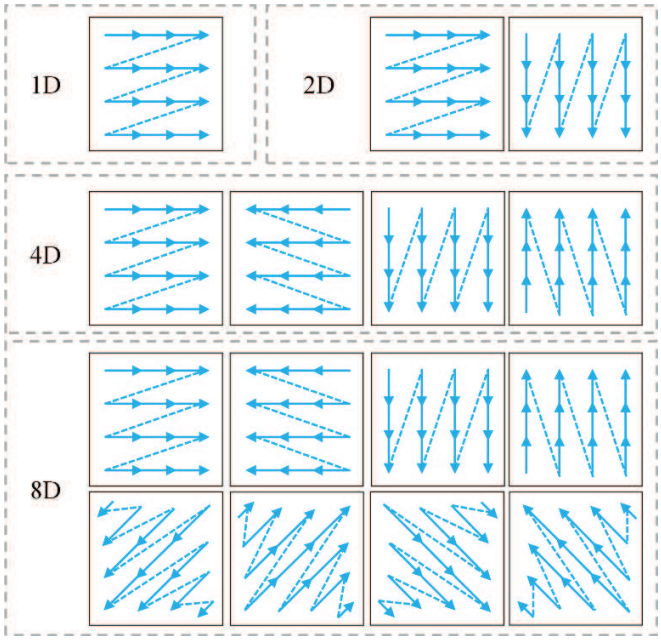


Fig. 9. Illustration of selective scanning directions for 1-D, 2-D, 4-D, and 8-D.

mIoU by 0.32% and 0.52%, respectively, over 1-D scanning. Compared to 2-D and 4-D variants, the 8-D configuration achieves at least 0.24% and 0.38% improvements in Mean $F1$ and mIoU, respectively. These findings indicate that incorporating global contextual information from multiple directions positively contributes to the semantic segmentation of remote sensing imagery, further validating the effectiveness of the eight-direction scanning mechanism.

TABLE VII
IMPACT OF SCANNING DIRECTION CONFIGURATIONS IN THE GLSS MODULE ON THE VAIHINGEN DATASET

Method	Evaluation metrics (%)		
	MeanF1	mIoU	OA
MF-Mamba-1D	86.39	76.40	87.97
MF-Mamba-2D	86.47	76.54	88.03
MF-Mamba-4D	86.43	76.47	87.87
MF-Mamba-8D	**86.71**	**76.92**	**88.30**

B. Generalization Capability

To assess the generalization capability of the proposed MF-Mamba model in remote sensing semantic segmentation tasks, we repartitioned the training and test sets of the ISPRS Potsdam dataset based on spatial distribution. Specifically, tiles 2_13, 2_14, 3_13, and 3_14 were used for training, while tiles 6_7, 6_8, 7_7, and 7_8 served as the test set. The selected training area is spatially far away from the test area, and the classes covered by the training area are limited. Thus, the features of some classes are not fully learned in the training set. Through this division strategy, we aim to further verify the generalization ability of the proposed model to nontraining areas. The results are presented in Table VIII.

MF-Mamba achieves the highest scores in terms of Mean F1, mIoU, and OA, outperforming the other methods. This superior performance is attributed to MF-Mamba's deep fusion

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TABLE VIII
GENERALIZATION PERFORMANCE ON THE SPATIALLY SEPARATED ISPRS POTSDAM SUBSET

Method	F1 (%)					Evaluation metrics (%)		
	Imp.suf	Building	Lowveg	Tree	Car	MeanF1	mIoU	OA
Deeplabv3+ [14]	83.25	94.83	81.12	78.19	87.58	84.99	74.36	84.31
ABCNet [35]	83.27	94.46	81.21	79.27	84.28	84.38	73.36	84.01
SegFormer [39]	83.71	94.30	82.03	79.67	86.01	85.14	74.48	84.76
UnetFormer [45]	83.43	94.23	81.17	79.38	86.42	84.93	74.17	83.94
CMTFNet [44]	84.60	**95.53**	82.47	79.69	87.83	86.02	75.89	85.42
MANet [37]	83.54	94.93	80.31	79.12	86.35	84.85	74.12	83.82
DDCMNet [55]	83.55	94.80	81.36	79.20	87.29	85.20	74.63	84.50
BANet [43]	82.90	94.46	81.83	79.32	85.50	84.80	73.99	84.37
RS3Mamba [52]	83.00	94.21	81.51	78.78	86.19	84.74	73.90	83.95
MF-Mamba	**84.71**	94.92	**83.34**	**79.94**	**87.99**	**86.18**	**76.08**	**85.45**

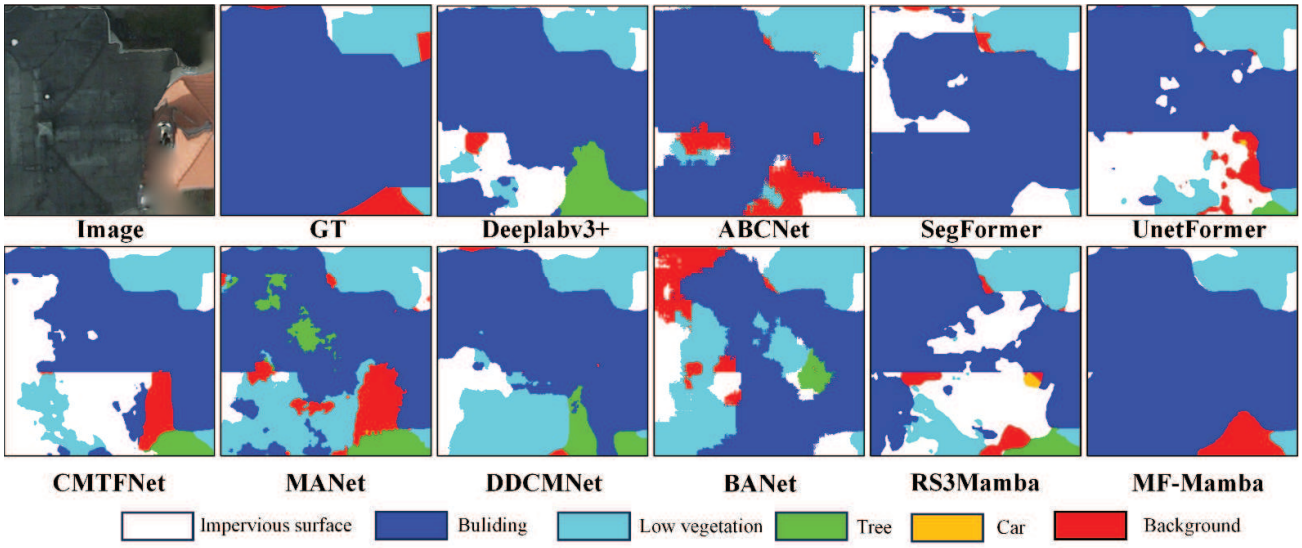


Fig. 10. Qualitative comparison results of generalization experiments.

and joint modeling of spatial, spectral, and scale-level features, which enables strong generalization even in test regions with differences in spatial pattern and class appearance from the training data. Fig. 10 presents the visual comparison results of specific scenes. In remote sensing images with complex backgrounds, where buildings share spectral or textural similarities with other classes (e.g., impervious surfaces or low vegetation), existing models often suffer from misclassification due to a lack of relevant training samples. In contrast, MF-Mamba can accurately capture the spatial structure and multiscale features of buildings and successfully distinguish them from confusing classes, further demonstrating the excellent generalization ability and robustness of MF-Mamba in complex scenes.

C. Model Complexity Analysis

Fig. 11 compares the computational complexity and parameter size of all models discussed in this study. The floating-point operations (FLOPs) are used to measure the inference-time computational load, while the parameter size represents the number of parameters of the model, which is an important indicator for measuring model complexity.

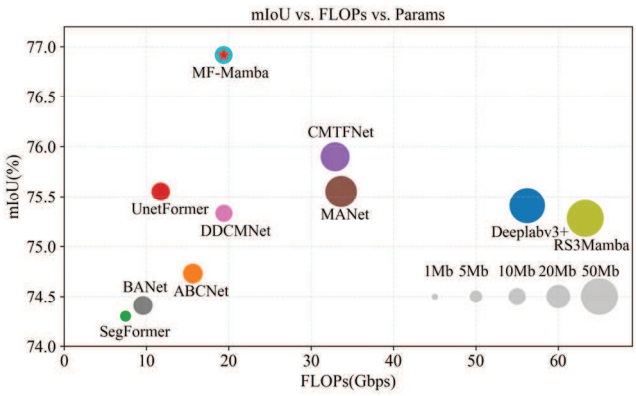


Fig. 11. Comparison of MF-Mamba with other methods in terms of parameter count, FLOPs (measured with an input size of 3 × 512 × 512), and mIoU on the ISPRS Vaihingen dataset. Circle sizes indicate the number of parameters.

As illustrated in Fig. 11, due to the application of deep separable convolution and a lightweight encoder, the number of parameters of MF-Mamba is only 10.91 Mb, which is significantly lower than that of the Transformer-based CMTFNet and CNN network MANet with a multiattention fusion strategy,

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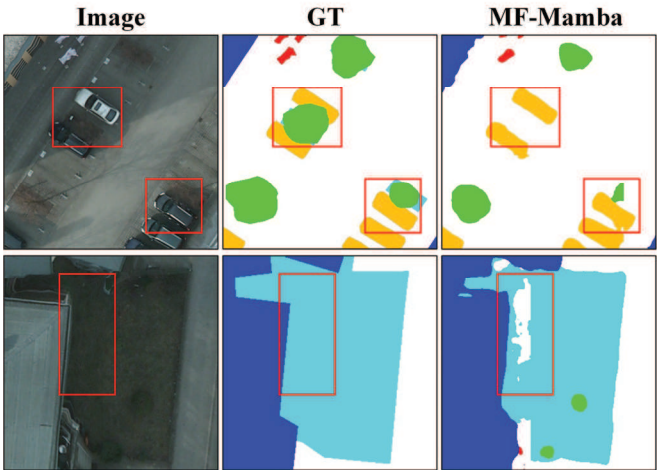


Fig. 12. Visualization of classification results with failure cases.

and it performs better in segmentation accuracy. In terms of computational complexity, relying on the efficient reasoning characteristics of the Mamba architecture, MF-Mamba has obvious advantages in computational efficiency compared to CMTFNet and MANet and is comparable to DDCMNet based on CNN. Although SegFormer integrates Transformer with a lightweight decoder to achieve the minimum parameters and computational complexity, this simple and lightweight design performs poorly when processing semantic segmentation tasks with multiple targets and scales in remote sensing images. In addition, ISPRS Vaihingen and ISPRS Potsdam are widely used as semantic segmentation benchmark datasets, and the optimized models for these two datasets can usually achieve high segmentation accuracy. On these datasets, MF-Mamba not only surpasses many advanced networks in accuracy but also maintains a low model complexity. In summary, the proposed MF-Mamba achieves superior segmentation performance with lower computational complexity.

D. Limitations and Future Research Directions

Although the proposed MF-Mamba model achieves excellent segmentation performance on multiple datasets, it still exhibits certain limitations. The appearance of 'shadows' caused by occlusion or light reflection is common in remote sensing images and often increases semantic ambiguity, thus hindering accurate segmentation. For example, as shown in Fig. 12, when low vegetation or trees exhibit weak spectral or textural features, they may appear similar to impervious surfaces under shadow interference, leading to misclassification. Addressing these issues is suggested as a key focus of future work.

Approaches for this could be as follows. First, the current model relies solely on single-modality optical imagery and lacks the ability to leverage multisource heterogeneous data. The growing availability of multimodal remote sensing data (e.g., DEM, SAR, and optical) presents opportunities for future research on cross-modal fusion [57] and attention-based interaction strategies [58] to improve the segmentation accuracy and scene adaptability. Second, the model's ability to capture complex spatial relationships remains limited. Integrating graph neural networks (GNNs) [59] may help reveal structured associations among ground objects and improve feature consistency in complex environments. Third, due to the scarcity of high-quality annotated samples in remote sensing, incorporating semi-supervised learning approaches [60] could enhance generalization under limited supervision. These directions offer promising pathways to address current challenges related to shadow interference, data diversity, spatial context modeling, and few-shot learning.

VI. Conclusion

This article proposed an MF-Mamba for remote sensing image semantic segmentation. MF-Mamba combines a CNN encoder and a Mamba-based decoder to extract global and local features. To leverage global and local information, we proposed the GLSS module within the decoder, which combines Mamba's global modeling capabilities with the CNN's local feature extraction power. To enhance the network's capability to represent features, we proposed the CSDF module to extract channel information, spatial information, and multiscale information in feature maps. Moreover, we designed multiscale lateral connections to effectively learn feature representations of different scales of the encoder.

We conducted extensive comparison and ablation experiments on the ISPRS Vaihingen, ISPRS Potsdam, and WHDLD datasets. The experimental results demonstrate that MF-Mamba outperforms existing models in segmentation accuracy, validating the effectiveness of the proposed model. Although outperforming other approaches, MF-Mamba still faces challenges in handling scenes with severe shadow interference or high visual similarity among thematic classes.

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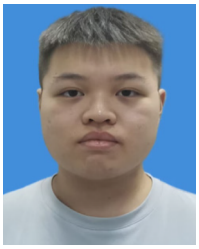
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